

Measuring Earth's Magnetic Field with a Smart Device – Teacher Copy

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In this class, students will use the [phyphox](#) app on a smart device to measure the strength and inclination of Earth's magnetic field at their school. This involves constructing a simple plumb bob to use alongside the app. They will take measurements both inside and outside, and compare their experimental results to the actual values from the current World Magnetic Model.

Earth & space science curriculum link:

Select and use equipment to generate and record data with precision, using digital tools as appropriate (AC9S7I03 AC9S8I03).

Pre-requisite knowledge:

- Students will understand that Earth has a dipolar magnetic field that can be quantified as a vector in terms of its strength and direction.
- They should be able to sketch a simple diagram Earth's magnetic field.
- They may have encountered the concepts of magnetic intensity and inclination before, otherwise these terms are defined in the background information for this class.
- Students should have basic familiarity with the phyphox app (see the document *Downloading and using the phyphox magnetometer*).

Aims:

By the end of this class, student should:

- Understand that the Earth's magnetic field is measured as 3 orthogonal components (x, y, z).
- Understand what a magnetometer is, and that smart devices like phones and tablets have a magnetometer sensor.
- Be able to use the phyphox app to collect geomagnetic data.

- Understand that Earth's magnetic field strength and inclination changes with latitude.
- Understand the concept of magnetic interference, and that electronic or metallic objects have their own magnetic fields that can interfere with geomagnetic readings.

Materials (Figure A):

- Mobile phone/ tablet with the [phyphox](#) app installed
- Protractor
- String/ twine (approx. 60 cm)
- Weight e.g., small shell/ pebble (use slotted weights if available)



Figure A: Materials for the Measuring Earth's Magnetic Field with a Smart Device class.

Safety considerations:

- Remind students to handle sharp craft objects carefully to avoid injury to other students/ staff.
- Handle smart devices with care to avoid damage.
- Carry out the outdoor component of this activity on a sports ground/ school yard away from roads and cars, and avoid uneven ground to minimise slips, trips and falls.

- Assess the weather conditions before taking students outside. If conditions are poor, you can carry out this class fully indoors.

Activity 1: Predicting the value of magnetic inclination at your location

Magnetic inclination is the angle that Earth's magnetic field lines make with Earth's surface.

A simple diagram showing how magnetic inclination changes with latitude is shown in **Figure 1**.

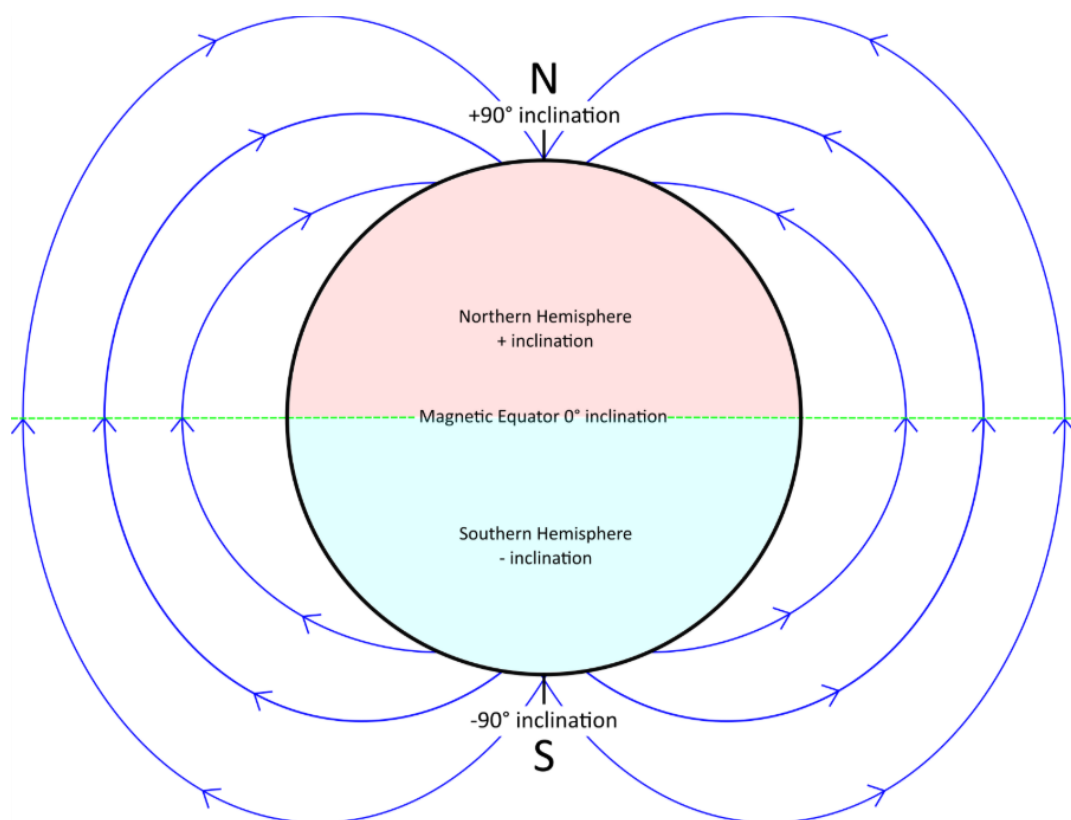


Figure 1: Simple diagram of Earth's magnetic inclination. Magnetic field lines and their direction are shown in blue.

1. What do you predict that the value of magnetic inclination will be at your location?

Students should predict the value of Earth's magnetic inclination based on the approximate latitude of their school.

In Hobart, Tasmania, magnetic inclination is approximately -72° (NCEI; BGS, 2024). Any approximation around -70° is correct for Tasmania.

Activity 2: Measuring magnetic field strength and inclination using phyphox

Split the class into small groups (2-3 students per group) with one smart device and set of equipment per group.

Step 1: Assembling a Plumb Bob to Measure Magnetic Inclination

Demonstration video: [Measuring Earth's Magnetic Field with a Smart Device Step 1: Assembling a Plumb Bob](#)

We will now construct a simple plumb bob that we can use to measure magnetic inclination on a smart device.

A **plumb bob** is simply a weight attached to a string. It is used as a vertical reference line to help measure angles.

1. Tie your weight e.g., a small pebble/ shell to one end of your string.
2. Tie the other end of your string around the short edge of your device so that there is approximately 20 cm of length hanging down.
3. Hold a protractor against the long edge of your device, with the plumb bob string against the outside of it.

Your set up should look like **Figure 2**.

We will use the angle that the string makes against the protractor to measure magnetic inclination.

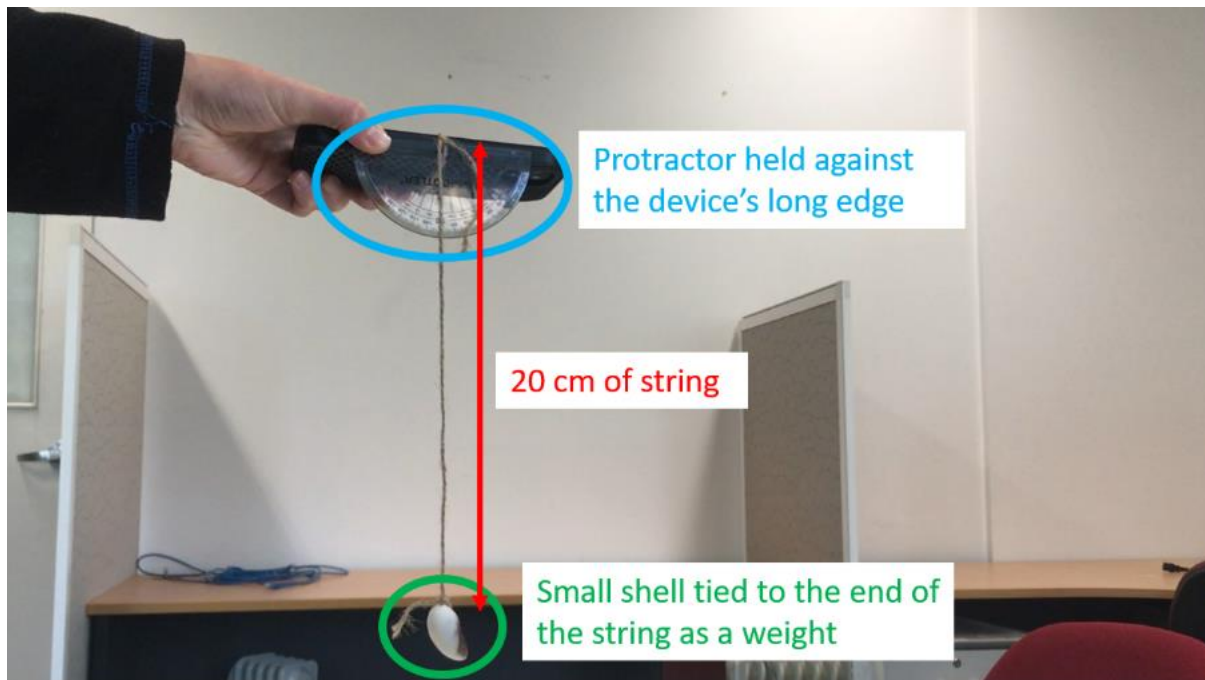


Figure 2: Labelled plumb bob set up.

Step 2: Measuring Magnetic Field Strength & Inclination

Demonstration video: [Measuring Earth's Magnetic Field with a Smart Device Step 2: Measuring Field Strength & Inclination](#)

We are now going to measure a value for Earth's magnetic field strength and an angle of magnetic inclination in your classroom.

Note: You will need to have the [phyphox](#) app downloaded on your device for this step. See the document '[Downloading and using the phyphox magnetometer](#)' for guidance.

1. Hold your device (including the plumb bob and protractor) horizontally out in front of you.
2. Start recording with the phyphox magnetometer in **Graph** mode.
3. Bring the value of **B_x to 0** by rotating your body whilst keeping your device flat (**Figure 3**).
4. When you have found the position where B_x is 0, keep your body still and bring **B_y to 0** by tilting your device vertically up/ down (**Figure 4**).
5. Hold your device in the position where B_x and B_y are both 0. Note what value **B_z** stabilises at – this is the magnetic field strength at your location. Record this in **Table 1**.

6. Also note what **angle** the plumb bob string makes against the protractor in this position – this is the angle of magnetic inclination at your location. Record this in **Table 1**.

Note: you may find it easier to get a partner/ member of your group to read off the angle of inclination whilst you hold everything still.



Figure 3: Bringing Bx to 0 by holding the device flat and turning your body.

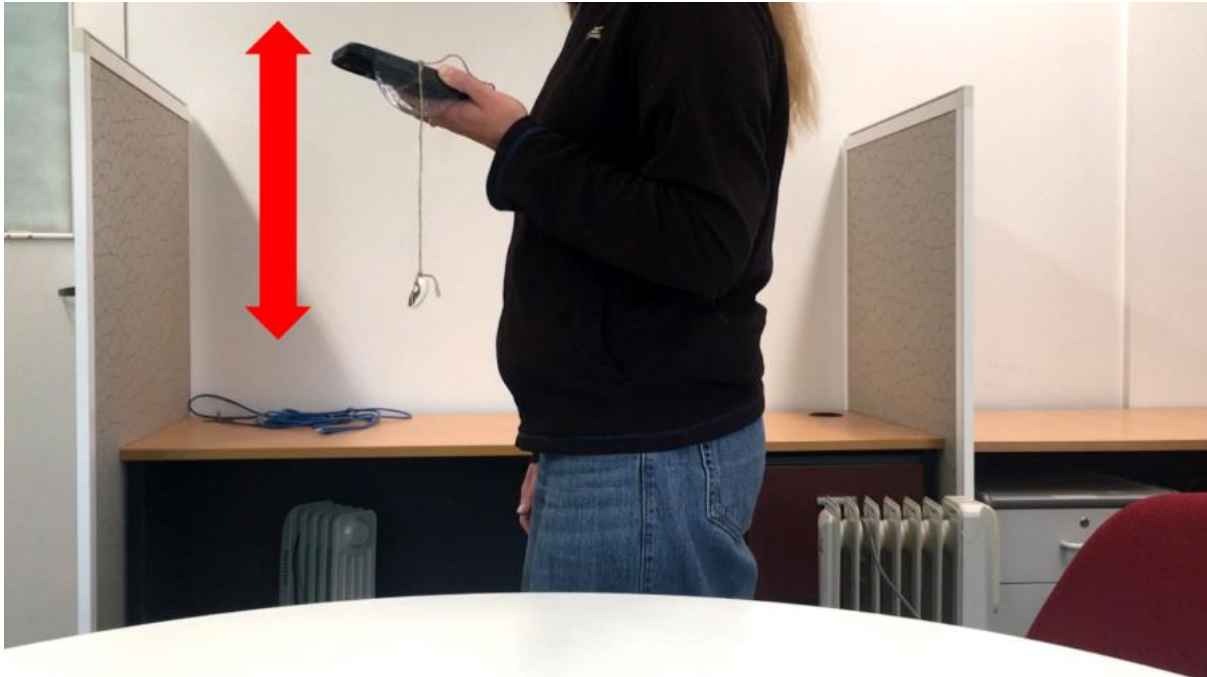


Figure 4: Bringing B_y to 0 by keeping your body still and tilting the device up/ down.

Note: You will notice that there are 2 opposite orientations you can face where B_x and B_y are both 0. You can measure the magnetic field from either direction.

If your device is tilted up, the magnetic inclination angle will be negative. If it is tilted down, the angle will be positive.

Students may need some assistance orientating their device to bring B_x and B_y to 0 correctly. See the [demonstration video](#) for guidance on how to do this.

An example app display where B_x and B_y have been brought to 0 is shown in **Figure B** below.

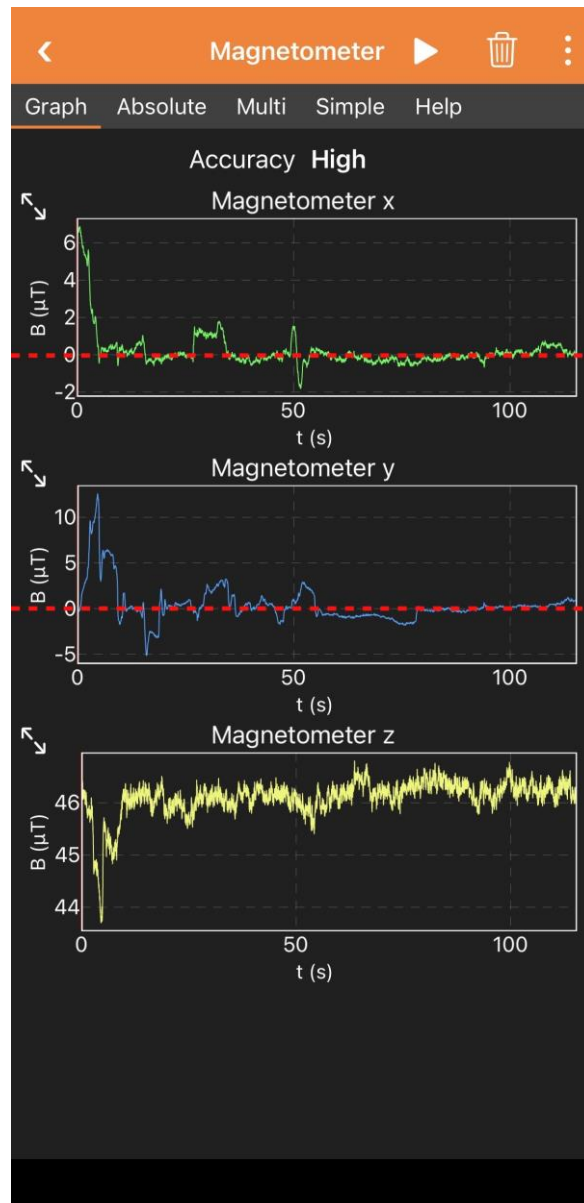


Figure B: Labelled screenshot of the phyphox magnetometer in **Graph** mode with B_x and B_y brought to 0. The red dashed lines mark the location of $0 \mu\text{T}$ for reference (phyphox app (RWTH Aachen University), used under [CC BY 4.0](#)).

B_z (the value of Earth's magnetic field strength) stabilises around $46 \mu\text{T}$ in this example.

Encourage students to work together in groups to read off the protractor value when measuring magnetic inclination. One student should hold the set up steady whilst another reads off the angle of the string against the protractor.

Extension question:

Why do you think we bring B_x and B_y to 0 to measure Earth's magnetic field?

We bring B_x and B_y to 0 so that the magnetic z axis is aligned perpendicular to your device i.e. the device is aligned with Earth's magnetic field line.

This is why there are 2 directions you can face where B_x and B_y are both 0 – you are aligned with the same field line but facing in opposite directions.

Note: this is a complex concept for students, and it is not essential for students to understand this bit of theory to complete the class successfully.

Your teacher will now take you outside onto your school field/ oval to repeat your measurements of magnetic field strength and inclination.

Record the values you collect outside in **Table 1**.

When taking students outside to repeat their measurements, choose a location away from buildings, moving cars, electronics, power lines, motors (e.g., air con units) and metal objects. A school playing field/ oval is ideal.

Activity 3: Comparing your results to the WMM

1. Is there a difference between the values of field strength and magnetic inclination you collected inside and outside?

Students should compare the readings for field strength and inclination that they collected inside the classroom and outside on the school grounds.

Their values of magnetic field strength will likely differ; however magnetic inclination values should be similar.

2. If there is a difference, what do you think caused this?

Encourage students to consider metallic or electronic objects in the classroom that may have influenced the results they collected inside. These objects have their own magnetic fields, and students may have measured these (as well as Earth's own magnetic field) if they took their measurements close by.

This is known as magnetic interference.

Figure 5 and **Figure 6** are maps of Main Field Total Intensity (F) and Main Field Inclination (I) from the **World Magnetic Model** (WMM).

The maps are also available on the WMM webpage [here](#) under the 'Maps' tab.

You may wish to print off large copies of the WMM intensity and inclination maps for the class to study.

What is the WMM?

The World Magnetic Model ([WMM](#)) is the standard model of the Earth's geomagnetic field. It is updated every 5 years to reflect any changes in the field. The model is developed by the British Geological Survey and the NCEI (NCEI; BGS, 2024).

Find your location on the WMM maps and read off the value for magnetic field intensity (aka strength) and magnetic inclination. Record these in **Table 1**.

Note: on the WMM maps, magnetic field intensity is given in nanoteslas (nT) whilst the phyphox app reports values in microtesla (μT).

To convert from nT to μT , divide the value by 1000. E.g., 50,000 nT = 50 μT .

1. How well do the values you collected compare to the values from the WMM?

Students should locate their approximate position on the WMM maps and record the values of field strength and inclination in **Table 1**. They should then compare their experimental values to the WMM values and see how similar they are.

If they followed instructions carefully, their results should be close to the WMM values. Their outside results should be more similar to the WMM than their inside results because there will be no/ less magnetic interference outside.

Remind students to convert the WMM values from nT to μT .

2. If they aren't a perfect match, can you think of any sources of error or uncertainties in your experimental set up?

Students should consider magnetic interference, human error (from reading off the protractor/ device screen) and calibration issues with the magnetometer in their device.

Table 1: Expected student results (in Hobart, Tasmania)

	Magnetic Field Strength (μT)	Magnetic Inclination ($^{\circ}$)
Inside	25 – 65 μT (Depending on proximity to electronic/ metallic objects which can cause magnetic interference)	Approx. 70° (Magnetic interference does not typically affect Earth’s magnetic inclination)
Outside	Approx. 60 μT	Approx. 70°
WMM values	62 μT (NCEI; BGS, 2024)	73° (NCEI; BGS, 2024)

US/UK World Magnetic Model - Epoch 2025.0 Main Field Total Intensity (F)

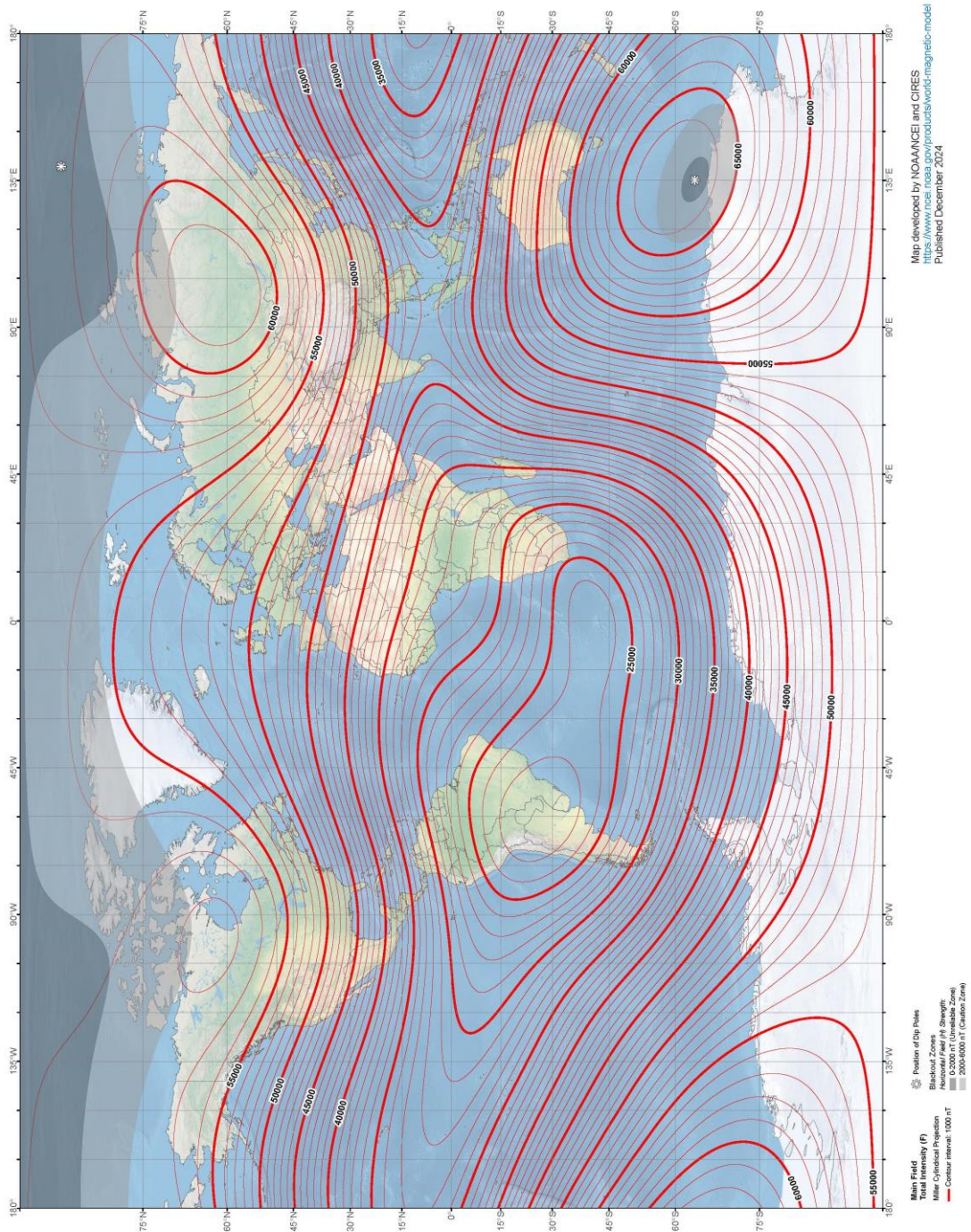


Figure 5: WMM map of magnetic field strength (NCEI; BGS, 2024).

US/UK World Magnetic Model - Epoch 2025.0
Main Field Inclination (I)

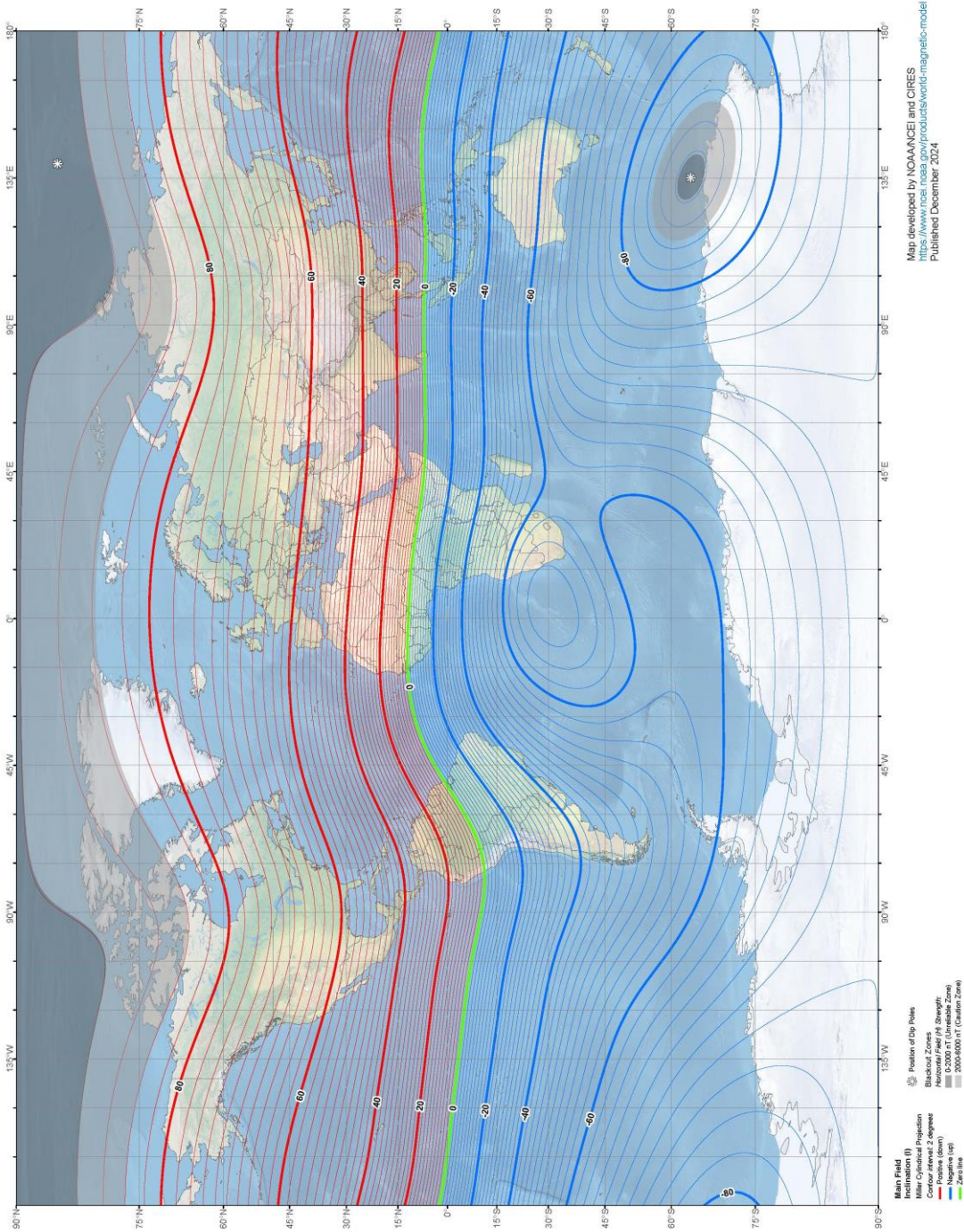


Figure 6: WMM map of magnetic field inclination (NCEI; BGS, 2024).

This activity was inspired by:

Labtastic (2023) Measure Earth's magnetic field with your smartphone. Available at:

https://youtu.be/AW8luI-GHSY?si=ZnreZb1jJzj_pVaN

NASA (2025). *Introductory Experiments in Magnetism with Smart Devices*. Available at:

<https://science.nasa.gov/learn/heat/resource/introductory-experiments-in-magnetism-with-smart-devices/>

NASA (2025). *Exploratory Activities in Magnetism with Smart Devices*. Available at:

<https://science.nasa.gov/learn/heat/resource/exploratory-experiments-in-magnetism-with-smart-devices/>

This activity uses the phyphox mobile app: phyphox (Physical Phone Experiments) app, 2nd Institute of Physics, RWTH Aachen University.

References:

GFZ German Research Centre for Geosciences (2025). *Frequently asked questions regarding geomagnetism*. Available at: <https://www.gfz.de/en/section/geomagnetism/data-products-services/frequently-asked-questions-regarding-geomagnetism#c14403>

NOAA NCEI Geomagnetic Modeling Team; British Geological Survey. 2024: World Magnetic Model 2025. NOAA National Centers for Environmental Information. <https://doi.org/10.25921/aqfd-sd83>